MILITARY SPECIFICATION

TEST SET, ALTIMETER SET AN/APN-347

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for shop use and shall provide signals and measuring devices necessary for the maintenance of the AN/APN-141 (V) Electronic Altimeter Set.

1.2 Classification - The test equipment covered by this specification shall consist of the following items:

Drawing/Model No.	Nomenclature	Ref. Para.
1529-0008	Test Set	3.5.1
104031-3	Case	3.5.5
104031-2	Cover	3.5.5.2

1.3 Associated Equipment - This equipment shall operate with the following associated equipment:

Test Equipment	Characteristics Required
Oscilloscope Tektronix 585A	5 inch screen, 4 traces, sweep speed .05 usec/cm to 50 usec/cm; trigger amplitude 0.5 to 60V pk/pk; vertical sensitivity: .02 v/cm x 5 magnification; frequency response: X-axis, DC to 1Mhz/sec; input impedance: 1 megohm on X-axis
Digital Voltmeter Cubic 240	Range: 0-40 VDC Accuracy: .01% or 1 digit
Electronic Counter Hewlett Packard HP524D	10 MHz Standard Frequency Stability 1 part in 10
Signal Generator Hewlett Packard 861A	Frequency: 4000-4500 MHz Power +4 DBM with a 100 db attenuator

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Voltmeter Electronic Hewlett Packard 410B	Ranges: 1V to 300V full scale in 6 ranges AC or DC. Accuracy: ±3% of full scale on all ranges for sinusoidal AC voltages at 400 Hz and for DC voltages.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-21200	Test equipment for use with Electronic and Fire Control Systems, General Specification for
MIL-T-18303	Test Procedure, Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment.
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
AR-5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of

PUBLICATIONS

Military

MIL-HDBK-217	Reliability Stress and Failure Rate Data for Electronic Equipment.

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PUBLICATIONS:

Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration; in so doing the following shall guide:

(a) Parts and materials requirements shall be in accordance with MIL-T-21200. Parts selected for use which are considered nonstandard per MIL-T-21200 shall be selected to meet the design requirements. Such parts shall only be used when a standard MIL equivalent part is not available or when delivery time of the standard part is beyond the time limit required to meet the contract delivery schedule.

(b) Semiconductor devices - Semiconductors shall be chosen and applied according to best engineering practice to attain high unit reliability and to meet all the design requirements.

3.2.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained in accordance with MIL-T-21200. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Electron Devices - Transistors and diodes shall be in accordance with MIL-T-21200.

3.2.3 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall be in accordance with AR-5

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of MIL-T-21200 for design, construction and workmanship except as otherwise specified herein.

3.3.1 Weight - The total weight of the equipment, excluding cables, shall not exceed 100 pounds.

3.3.2 Reliability

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3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 150 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 25,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.3.2.3 Reliability in Meantime Between Failure - The predicted mean (operating) time between failure in accordance with MIL-HDBK-217 shall be 5,000 hours minimum.

3.3.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.3.3.1 Cable Assemblies, Power Electrical - Applicable input power cable and connections shall conform to MIL-T-21200.

3.3.4 Control Panels - All rack mounted control panels shall conform to the applicable requirements as specified in MIL-T-21200.

3.3.5 Interchangeability - The equipment shall meet the applicable interchangeability requirements as specified in MIL-T-21200.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-T-21200.

3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

(a)	Temperature	Room ambient (25°C ±5°C)
(b)	Altitude	Normal Ground
(c)	Vibration	None
(d)	Humidity	Room ambient up to 90% relative humidity
(e)	Input power voltage	115 ± 1.0 VAC, 400 ± 5Hz

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3.3.10 Service Conditions - The equipment shall operate satisfactorily when subjected to the environmental service conditions in accordance with MIL-T-21200 for Class 3 equipment except as modified herein.

(a) Vibration

(b) Temperature/Altitude

(c) Shock

(d) Humidity

3.3.11 Warm-up Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 10 minutes at extreme service conditions.

3.3.12 Primary Input Power Requirements - The test equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in MIL-T-21200. The power required shall be of the following type and shall not exceed the specified amounts.

AC Power, Single Phase, 115V, 380-420 Hz, 400VA

3.3.12.1 Protection - The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in MIL-STD-704.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Test Capabilities - To be provided as follows:

(a) System sensitivity and accuracy test - The equipment shall provide for overall AN/APN-141 (V) system test and alignment.

(b) Receiver accuracy test - A test capability shall be provided for alignment and accuracy test of the receiver portion of the Receiver Transmitter unit.

(c) Fault Isolation - This test capability shall provide for fault isolation at the Altimeter Height Indicator, Coupler, Receiver-Transmitter and RF Switch units.

(d) Adjustable Voltage - The test set shall provide by means of an adjustable transformer, a 400 Hz voltage having a range of zero to 130 volts.

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3.5 Detail Requirements

3.5.1 System Sensitivity and Accuracy Test Characteristics -- The test set shall receive the transmitted 4300 MHz signal of the AN/APN-141 (V) system via a coaxial cable from the transmitter output. It shall process the signal and return it to the receiver via coaxial cable.

The signal shall be processed to accurately simulate a return signal in the high altitude mode at 1000 feet and in the low altitude mode at 0 feet, 10 feet, and 40 feet.

The signal shall be attenuated by means of fixed and variable attenuators to allow for sensitivity measurements.

The Altimeter transmitter frequency shall be measured by monitoring the detected output of a cavity wave meter.

A signal sampler shall be provided for making relative transmitter power measurements.

3.5.1.1 Accuracies shall be.

Parameter	Tolerance
Delay	+ 2% or + 4 ft. whichever If greater
Sensitivity	+ 5db
Frequency Measurement	± 10 MC
RF Power Measurement	± 6 db

3.5.2 Receiver Accuracy Test Characteristics - Circuitry shall be provided to convert a system signal to a simulated ground return pulse. The pulse delay shall be programmed to provide fixed altitude delays of 100 feet, 150 feet, 300 feet, 1000 feet, 3000 feet and 5000 feet

Circuitry shall also be provided for programming the signals required to test the altimeters tracking rate in the high and low altitude modes

A dummy load shall be provided to load the transmitted signal while performing the receiver accuracy tests It shall have an impedance of 50 OHMS

3.5.2.1 Simulated Signal Accuracy Characteristics - The simulated altitude signals shall have a delay accuracy equivalent to an altitude of ± 4 feet or 2%.

3.5.3 Fault Isolation Tests Characteristics - The test set shall provide a variable dc signal capable of simulating the Input Analog Signal to the Altimeter Indicator, Altitude signal to the Coupler, and Range voltage to the RF Switching Unit

Test jacks shall be provided to monitor input and output signal lines for all units of the altimeter system.

A voltage divider shall be provided to scale the Coupler Altitude signal out to a voltage representing a direct altitude reading.

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3.5.3.1 DC Signal Accuracy:

Range	Tolerance
1V to 21V	Continuously variable Low limit -0.5 volts High limit +26 volts

3.5.3.2 Voltage Divider Scale Accuracy:

Scale Factor	Tolerance
1.00V=1000 feet	± 5%

3.5.4 Form Factor - The system test set maximum dimensions shall be twenty (20) inches deep, twenty two (22) inches wide and seventeen (17) inches high. The case assembly being eleven (11) inches high and the cover assembly being six (6) inches high.

3.5.4.1 Weight - The weight of the system test set shall not exceed one hundred pounds.

3.5.4.2 Cover and Case Assembly - The equipment shall be housed in a combination case, composed of two sections, the Cover Assembly and the Case Assembly. It shall have the capability of being separated.

3.5.4.3 Cover Assembly - The cover assembly shall have a hinged inner lid and shall be used to store and transport the frequency meter, heat sink assembly and eleven cable assemblies.

3.5.4.4 Case Assembly - The case assembly shall have a removable electronic panel assembly containing.

(a) Power sources
(b) Microwave components
(c) Electronic sub-assemblies and components

3.5.4.4.1 Feet - The case assembly shall have two sets of feet, one set on the bottom and one set on an end as provision for resting the case on a bench.

3.5.4.4.2 Handles and Latches - The cover assembly and the case assembly shall each have two handles. Latches shall be provided for attaching the cover to the case.

3.5.4.4.3 Heatsink Plate and Associated Brackets - Supplied with the equipment shall be a heatsink plate and associated brackets to facilitate heat dissipation and mounting of the altimeter set under test and a bracket to support the frequency wave meter.

3.5.5 Contents - The Test Set AN/APM-347 shall contain the following assemblies/subassemblies having the indicated characteristics.

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Assemblies/Subassemblies	Characteristics
(1) Altitude Signal Simulator Card	Provide simulated altitude pulsed outputs in six steps from 100 to 5000 feet with an accuracy of 2%.
(2) Drive Circuit Assembly	Primary purpose is to shape, amplify and bias the input pulse from the altitude signal generator and to drive the bias port of the crystal switch.
(3) +150 Volt Power Supply	Input voltage range: 105-125 volts Input frequency: 50-400 Hz Regulation--Line: ± 0.05% Ripple: 0.004% or 3MVRMS whichever is greater with 400 Hz Output. 144.0 to 160.0 volts, adjustable.
(4) +34 Volt Power Supply	Input voltage range: 105-125 volts Input frequency. 50-400 Hz Regulation—Line: ± 0.5% Ripple 0.04% or 10 MVRMS whichever is greater than 400 Hz.
(5) +12 Volt Regulator	Input voltage range 16-35 VDC Output: 12.0 VDC nominal Output adjustment range 11.4-12.6 VDC Regulation: ± 0.5% for 30% input change.
(6) +4.7 Volt Regulated Supply	Input voltage +12 VDC Output. +4.7 VDC Regulation: ± 5%
(7) -5 Volt Regulated Supply	Input voltage 115 VAC, 400 Hz Output -5 VDC Regulation: ± 5%
(8) Electronic Coaxial Switch.	Provides remote rf switching Operating voltage. 115 VAC RF output positions: 6 Switching rate: 30 operations/sec Switching time 50 msec/60°throw

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Assemblies/Subassemblies	Characteristics
	Operating frequency: 0-12 GHz VSWR: DC to 8 GHz 1.3 max. Insertion loss: DC to 5 GHz 0.3 db Isolation: DC to 12 GHz 70 db aver. Average power capability: DC to 10 GHz 400W RF voltage rating: 1500 volts.
(9) 10 & 40 Foot Delay Lines	Simulates 10 foot altitude delay path consisting of precisely measured length of coaxial transmission line. Delay: 20 nanoseconds, ± 2%. Simulated 40 foot altitude delay path consisting of precisely measured length of coaxial transmission line. Delay: 80 nanoseconds ± 2%.
(10) 1000 Foot Delay Line	Simulates 1000 foot altitude delay path Type: Acoustical delay line Frequency: 4.3 GHz Insertion loss 35 db Delay: 2.02 usec ± 2% Spurious response signals including triple travel delayed signal level, minimum of 20 db VSWR: 4.1 max. Input power peak 2 watts max average 100%.
(11) Crystal Switch	Shall pass RF pulse when gated with resultant envelope of rise time of better than 2 db/nsec and fall time of better than 2 db/nsec.

3.5.6 Controls - The equipment controls, their functions and location are as follows:

3.5.6.1 System Test Controls

Control Name	Function	Location
Attenuator	Sets RF Signal Level	Front Panel
ALT SEL	Selects System Delays or Transmitter load	Front Panel
Cavity Wavemeter	Tunes to transmitted Frequency	Accessory Panel

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3.5.6.2 Receiver Accuracy Test Controls

3.5.6.2.1 External Controls

Control Name	Function	Location
Alt Feet Program Selector	Selects fixed or variable altitude steps	Front Panel
PLS Width	Adjust Modulating Pulse Width	Front Panel
Mode	Selects Altitude Program or rate	Front Panel
Alt Rate CTR	Adjusts High or Low rate center	Front Panel
Alt Sig	Adjust variable altitude from 0 to 5000 feet	Front Panel

3.5.6.2.2 Internal Controls

Control Name	Reference Designation	Function	Location
Ladder Adjust	R30	100' Adjust	A-1 Board
Ladder Adjust	R29	150' Adjust	A-1 Board
Ladder Adjust	R28	300' Adjust	A-1 Board
Ladder Adjust	R27	1000' Adjust	A-1 Board
Ladder Adjust	R26	3000' Adjust	A-1 Board
Ladder Adjust	R25	5000' Adjust	A-1 Board
Ladder Adjust	R24	Rate Adjust	A-1 Board
Level Adjust	R31	Adjusts High Rate Level	A-1 Board
Level Adjust	R32	Adjusts Low Rate Level	A-1 Board
Rate Swing Adjust	R21	Adjusts Low Rate Swing	A-1 Board
Rate Swing Adjust	R22	Adjusts High Rate Swing	A-1 Board
Threshold Swing Adjust	R3	Adjusts Threshold Level	A-1 Board

3.5.6.2.3 Fault Isolation Test Controls

Control Name	Function	Location
+12 Volt Adjust	12 Volt Power Supply Adjust	PS-3 Supply
+150 Volt Adjust	150 Volt Power Supply Adjust	PS-1 Supply
+34 Volt Adjust	34 Volt Power Supply Adjust	PS-2 Supply
115 VAC Adjust	115 VAC Input Adjust	Front Panel
Test Sel	Selects System, Test, Indicator Test, Coupler Test, or Manual Program	Front Panel
Reliability	Selects simulated reliable or unreliable signal or system reliability signal	Front Panel
115 VAC Power Switch	Turns 115 VAC Power ON or OFF	Front Panel

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3.5.6.3 System Test Set Indicators

Indicator Name	Function	Location
Lo Alt Lim	Indicates low altitude limit	Front Panel
Self Test	Indicates self test	Front Panel
115 VAC	Indicates 115 VAC Power ON	Front Panel
Sys Current	Indicates system current drain	Front Panel
Sys Voltage	Indicates input voltage setting	Front Panel

3.5.7 Electrical Connections

Reference Designation	Receptacle Type	Function
J1	MS 3110E-16-26S	To Indicator
J2	MS 3110E-12-8S	To RF Switch
J3	MS 3110E-16-26SW	To Receiver/Transmitter
J4	MS 3110E-14-15S	To Coupler
J5	UG 30D/U	RF Mon
J6	T 3001-75	To RCVR
J7	UG 30D/U	4.3 GHZ input
J8	UG 492 D/U	XMTR Video
J9	T 3001-7[illegible]	From XMTR
J10	MS 3102E-145-7P	115 VAC Input Power

3.5.8 Case, Test Set AN/APM-347 - The test set shall be provided with a combination case which shall conform to MIL-T-21200.

3.5.8.1 Size - Twenty (20) inches deep, twenty-two (22) inches wide and eleven inches high.

3.5.8.2 Cover, Test Set AN/APM-347

3.5.8.2.1 Size - Twenty (20) inches deep, twenty-two (22) inches wide and six (6) inches high.

3.5.8.3 Material - Aluminum Alloy 6061-0

3.5.8.4 Finish - Paint, light gray, semigloss enamel Color No. 26307 per MIL-STD-595.

3.5.8.5 Stowage - Sufficient stowage volume shall be provided for the items listed in para. 3.5.9.

3.5.9 Cable Assemblies and Accessories - Cable assemblies and accessories shall be provided as follows

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Drawing No.	Nomenclature
1529-2301	Cable Assy
1529-2302	Cable Assy
1529-2303	Cable Assy
1529-2304	Cable Assy
1529-2305	Cable Assy
1529-2306	Cable Assy
1529-2307	Cable Assy
1529-2308	Cable Assy
1529-2309	Cable Assy
1529-2310	Cable Assy
1529-2311	Cable Assy
1529-2312	Cable Assy
	Instruction Handbook
	Bag Containing:
1529-1110	Purging Adapter
NAS1515D061	Washer
M24129/3-3P	Nut
MS51958-64	Screw
MS51958-68	Screw

4 QUALITY ASSURANCE

4.1 Responsibility for Inspection - Unless otherwise specified in the contract, or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections deemed necessary to assure that supplies and services conform to prescribed requirements

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(a) Preproduction Tests
(b) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.4 The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the test when so desired Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature

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shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction test shall include environmental tests in accordance with the requirements of MIL-T-5422 to the limits specified in MIL-T-21200.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Acceptance Tests - The supplier shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Suppliers not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The supplier shall furnish test reports showing the quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the supplier. Acceptance test shall consist of individual unit tests.

(1) Individual Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability specified herein. As a minimum each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4 Test Procedures - The procedures used for conducting the preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector, to modify or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Test specifications shall be prepared in accordance with MIL-T-18303. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.5 Presubmission Testing - No item, part or complete equipment shall be submitted until it has been previously tested and inspected by the supplier and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.5.1 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejections and the action taken to correct the defects found in the original shall be furnished to the Government Inspector.

5 PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - This equipment is intended to provide the test capability required for fault isolation, repair and adjustment of the AN/APX-111 (V) Transponder, down to the module level.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than as specified in MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.2.1 As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplied), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.3 Precedence of Documents - When the requirement of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity

(c) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications shall apply to the extent specified.

6.4 Ordering Data - Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following

(1) Title, number, and date of this specification
(2) Selection of applicable levels of packaging and packing (See 5.1)

6.5 Definition - The following non-military definition shall apply throughout this specification.

6.5.2 Supplier - The facility fabricating, assembling, acceptance testing and furnishing the equipment in accordance with the contract specifications

Project No. 6625-1501

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

MIL-T-81831(AS) TEST SET, ALTIMETER SET AN/APM-347

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

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Naval Air Systems Command
Washington, D.C. 20360

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